

Task Tracking Application Bug Fixes.

Bug #1 → SQL WHERE clause precedence bug.

* Where

→ TaskRepository.java, the @Query string in WHERE clause
→ exact same bug also found in following 2 files.

→ db/queries/search-tasks.sql

→ db/oracle/task-search-package.sql.

* How I found it

→ while testing the app, I found that even after changing the status filter in dropdown, it didn't actually filter.
ie for "IN PROGRESS" status also I was getting all tasks.

→ Also while testing the search API with curl, I saw archived tasks showing up too, even though they are supposed to be hidden.

* Root cause

→ In SQL, AND is evaluated before OR like in ~~math~~ mathematics * is evaluated before +.

→ The query had no brackets so it was actually being read as two separate ~~parts~~ halves joined by OR

WHERE archived = FALSE AND LOWER(title) LIKE :term
↳ group 1

OR

LOWER(description) LIKE :term AND (:status IS NULL OR status = :status)

↳ group 2

→ In group-1, status never appears in group 1 at all.
it only checks archived=FALSE and title LIKE term.

→ When search box is empty, term becomes %%. and %%. matches everything (even empty text)

→ So ~~searched~~ group 1 basically becomes just archived = FALSE
ie true for almost every Non archive task.

→ Since its group 1 OR group 2; if group 1 is already true the row gets included no matter ~~what~~ what group 2 says.

→ So status check, mentioned in group 2 gets ignored

→ That's why selecting any status in the dropdown, shows all tasks.

Now why archived tasks leaked in search results?

→ group 2 only checks description LIKE term and the status filter. It never checks archived at all

→ So if task's description matches the search word, it gets included through group 2

→ It doesn't matter if that task is archived, group 2 has no idea about the archived flag. The archived check only exists in group 1, not in group 2.

→ So an archived task gets included into result, just by matching description.

* **Fix**

→ added brackets around title / description OR part
so all three conditions apply together,

→ did same fix all following 3 files.

→ primarily in `TaskRepository.java.`

→ in `db/queries/search-tasks.sql`

→ in `db/oracle/task-search-package.sql`